

JEE MAIN REVISION MASTER: HYPERBOLA SYLLABUS REDUCED

Subject: Mathematics | Target Scope: JEE Main Previous Years | Empirical Base: 95 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Hyperbola - Core Application	95 PYQs	General Concept

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
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3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Hyperbola - Core Application (Very High Yield — 95 PYQs)

Core Principle: Key problem pattern in Hyperbola.

■ ■ Caution / JEE Pitfall: Watch for algebraic and sign errors.

■ Speed Trick: Verify dimensions and boundary conditions.

Representative Source PYQs:

- [2026 Evening Shift] Q#1 (MCQ): "The eccentricity of an ellipse E with centre at the origin O is $\frac{\sqrt{3}}{2}$ and its directrices are $x = \pm \frac{4\sqrt{6}}{3}$..." [Ans: D]
- [2026 Evening Shift] Q#1 (MCQ): "Let $H: \frac{x^2}{2a^2} - \frac{y^2}{2b^2} = 1$ be a hyperbola such that the distance between its foci is 6 and the distance between its d..." [Ans: B]
- [2026 Morning Shift] Q#1 (MCQ): "Let the domain of the function $f(x) = \log_3 \log_5 \log_7 (9x - x^2 - 13)$ be the interval (m, n) . Let the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: JEE Main 2026 (Online) 6th April Evening Shift (Q#1)

The eccentricity of an ellipse E with centre at the origin O is $\frac{\sqrt{3}}{2}$ and its directrices are $x = \pm \frac{4\sqrt{6}}{3}$. Let $H: \frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ be a hyperbola whose eccentricity is equal to the length of semi-major axis of E , and whose length of latus rectum is equal to the length of minor axis of E . Then the distance between the foci of H is :

- (A) $\frac{4\sqrt{2}}{\sqrt{7}}$
- (B) $\frac{4\sqrt{2}}{7}$
- (C) $\frac{4\sqrt{7}}{7}$
- (D) $(8/\sqrt{7})$ [Correct]

Correct Answer: D

Solution Analysis: For the ellipse E , let its equation be $\frac{x^2}{A^2} + \frac{y^2}{B^2} = 1, A > B$ since its major axis is along the x -axis because the directrices are of the form $x = \text{constant}$. Given eccentricity, $e = \frac{\sqrt{3}}{2}$. For an ellipse with major axis along the x -axis, the directrices are $x = \pm (A / e)$. We are given $x = \pm \frac{4\sqrt{6}}{3}$. So, $(A / e) = \frac{4\sqrt{6}}{3}$. Substituting $e = \frac{\sqrt{3}}{2}$, $A = \frac{\sqrt{3}}{2} \cdot \frac{4\sqrt{6}}{3} = \frac{4\sqrt{18}}{6} = \frac{4 \cdot 3\sqrt{2}}{6} = 2\sqrt{2}$. Thus the semi-major axis of ellipse E